CAPSTONE – 1: AI-Based Solar Power Generation Prediction System

1. Project Question — What do you have to do?

Develop an ML-based system that predicts short-term solar power generation using historical power output and environmental measurements. The system should help users understand how sunlight and weather conditions affect solar energy production.

- Collect a suitable solar-generation dataset or record panel output alongside sunlight intensity, temperature, and time.
- Clean the data and perform Exploratory Data Analysis (EDA) to identify daily and seasonal patterns.
- Create useful features such as hour, day, irradiance, temperature, and previous power readings.
- Train a regression model to predict solar power output for a defined future interval.
- Display actual and predicted generation, trends, and prediction errors on a dashboard.
- Compare at least two suitable models and evaluate them on a time-ordered test set.

2. Required AI Skills

AI Skill	Requirement
Python programming	Basic data handling and model development
Data preprocessing	Missing values, outliers, scaling where needed
EDA	Solar output and weather patterns
Feature engineering	Time, irradiance, temperature, and historical output
Machine learning	Linear Regression, Random Forest Regressor, or Gradient Boosting
Model evaluation	MAE, RMSE, and R^2
Visualization	Actual-versus-predicted power graphs
Model integration	Connect predictions to a dashboard

3. Required Tools and Technologies

Tool	Purpose
Python	Main programming language
Google Colab / Jupyter	Analysis and model training
Pandas, NumPy	Data processing
Matplotlib, Seaborn	EDA and visualization
Scikit-learn	Regression models and evaluation
Streamlit	Interactive dashboard
Public solar dataset	Historical generation and weather data
Solar panel and sensors (optional)	Small-scale experimental data collection

4. Final Submission Requirements

- Complete source code and notebook
- Dataset source link and feature descriptions

- Preprocessing steps and EDA graphs
- Trained model and algorithm explanation
- MAE, RMSE, R^2 and comparison results
- Dashboard showing actual and predicted solar output
- Sample predictions and screenshots
- Project report with architecture, methodology, limitations, and results
- Final presentation (PPT) and viva demonstration
- Solar panel measurement demonstration, if hardware is included